Practical No. 1

Aim: Write a java program for basic arithmetic operations.

Code:

Program 1:

```
public class BasicArithmeticStatic {
   public static void main(String[] args) {
      // Predefined numbers
      int num1 = 25;
      int num2 = 5;

      System.out.println("--- Basic Arithmetic (Static Values) ---");
      System.out.println("Number 1: " + num1);
      System.out.println("Number 2: " + num2);

      // Operations
      System.out.println("Addition (+): " + (num1 + num2));
      System.out.println("Subtraction (-): " + (num1 - num2));
      System.out.println("Multiplication (*): " + (num1 * num2));
      System.out.println("Division (/): " + (num1 / num2));
      System.out.println("Modulus (%): " + (num1 % num2));
   }
}
```

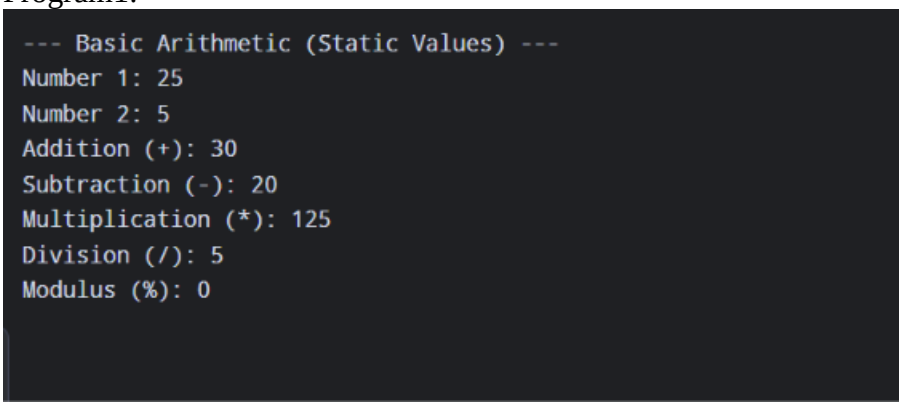
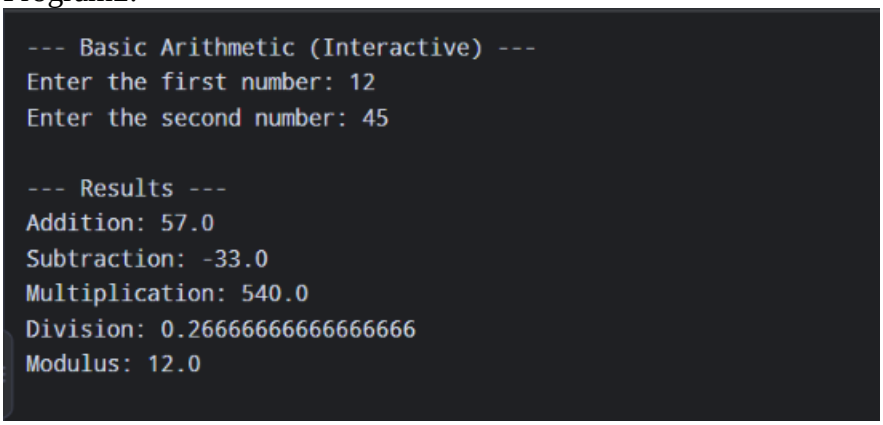
Program 2:

```
import java.util.Scanner;
public class BasicArithmeticInteractive {
   public static void main(String[] args) {
      Scanner scanner = new Scanner(System.in);

      System.out.println("--- Basic Arithmetic (Interactive) ---");

      // Take inputs
      System.out.print("Enter the first number: ");
      double num1 = scanner.nextDouble();

      System.out.print("Enter the second number: ");
      double num2 = scanner.nextDouble();

      // Perform operations and display results
      System.out.println("\n--- Results ---");
      System.out.println("Addition: " + (num1 + num2));
      System.out.println("Subtraction: " + (num1 - num2));
      System.out.println("Multiplication: " + (num1 * num2));

      // Handling division by zero safeguard
```

```
        if (num2 != 0) {
            System.out.println("Division: " + (num1 / num2));
            System.out.println("Modulus: " + (num1 % num2));
        } else {
            System.out.println("Division & Modulus: Cannot divide by zero.");
        }

        scanner.close();
    }
}
```

Output:

Program1:



Program2: